

HDOC C Programming

Variable-Length 2D Arrays

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Question 1

Read positive row and column values, declare a variable-length 2D integer array, read elements column by column, and print them in row-major order using one loop and a linear index.

C Program

```
#include <stdio.h>

int main()
{
    int rows, cols;

    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &rows);
        if (rows <= 0)
            printf("Rows must be positive.\n");
    } while (rows <= 0);

    do
    {
        printf("Enter number of columns: ");
        scanf("%d", &cols);
        if (cols <= 0)
            printf("Columns must be positive.\n");
    } while (cols <= 0);

    int arr[rows][cols];

    printf("\nEnter array elements column by column:\n");

    for (int j = 0; j < cols; j++)
    {
        for (int i = 0; i < rows; i++)
        {
            printf("Enter element [%d][%d]: ", i, j);
            scanf("%d", &arr[i][j]);
        }
    }

    printf("\nArray elements in row-major order:\n");

    for (int k = 0; k < rows * cols; k++)
    {
        int i = k / cols;
        int j = k % cols;
        printf("%d ", arr[i][j]);

        if ((k + 1) % cols == 0)
            printf("\n");
    }

    return 0;
}
```

Question 2

Read elements into a variable-length 2D array, copy the elements into a 1D array in row-major order, and display both arrays.

C Program

```
#include <stdio.h>

int main()
{
    int rows, cols;

    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &rows);
        if (rows <= 0)
            printf("Rows must be positive.\n");
    } while (rows <= 0);

    do
    {
        printf("Enter number of columns: ");
        scanf("%d", &cols);
        if (cols <= 0)
            printf("Columns must be positive.\n");
    } while (cols <= 0);

    int arr[rows][cols];
    int oneD[rows * cols];

    printf("\nEnter elements of the 2D array:\n");

    for (int i = 0; i < rows; i++)
    {
        for (int j = 0; j < cols; j++)
        {
            printf("Enter element [%d][%d]: ", i, j);
            scanf("%d", &arr[i][j]);
        }
    }

    for (int i = 0; i < rows; i++)
    {
        for (int j = 0; j < cols; j++)
            oneD[i * cols + j] = arr[i][j];
    }

    printf("\n2D Array:\n");

    for (int i = 0; i < rows; i++)
    {
        for (int j = 0; j < cols; j++)
            printf("%d ", arr[i][j]);
        printf("\n");
    }

    printf("\n1D Array in row-major order:\n");

    for (int i = 0; i < rows * cols; i++)
        printf("%d ", oneD[i]);

    printf("\n");
    return 0;
}
```

Important Row-Major Mapping

```
oneD[i * cols + j] = arr[i][j];
```

This maps the element at row *i* and column *j* of the 2D array to its corresponding linear position in the 1D array.

Execution Screenshots

Question 1 - Variable-Length 2D Array: column-wise input and row-major output

```
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day28.1.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day28.1.c -o day28.1
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day28.1
Enter number of rows: 2
Enter number of columns: 3

Enter array elements column by column:
Enter element [0][0]: 1
Enter element [1][0]: 4
Enter element [0][1]: 2
Enter element [1][1]: 5
Enter element [0][2]: 3
Enter element [1][2]: 6

Array elements in row-major order:
1 2 3
4 5 6
dhruvpandey@DHRUVs-MacBook-Air ~ %
```

Additional execution test

```
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day28.1.c -o day28.1
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day28.2
zsh: no such file or directory: ./day28.2
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day28.1
Enter number of rows: 3
Enter number of columns: 3

Enter array elements column by column:
Enter element [0][0]: 10
Enter element [1][0]: 20
Enter element [2][0]: 30
Enter element [0][1]: 40
Enter element [1][1]: 50
Enter element [2][1]: 60
Enter element [0][2]: 70
Enter element [1][2]: 80
Enter element [2][2]: 90

Array elements in row-major order:
10 40 70
20 50 80
30 60 90
dhruvpandey@DHRUVs-MacBook-Air ~ %
```